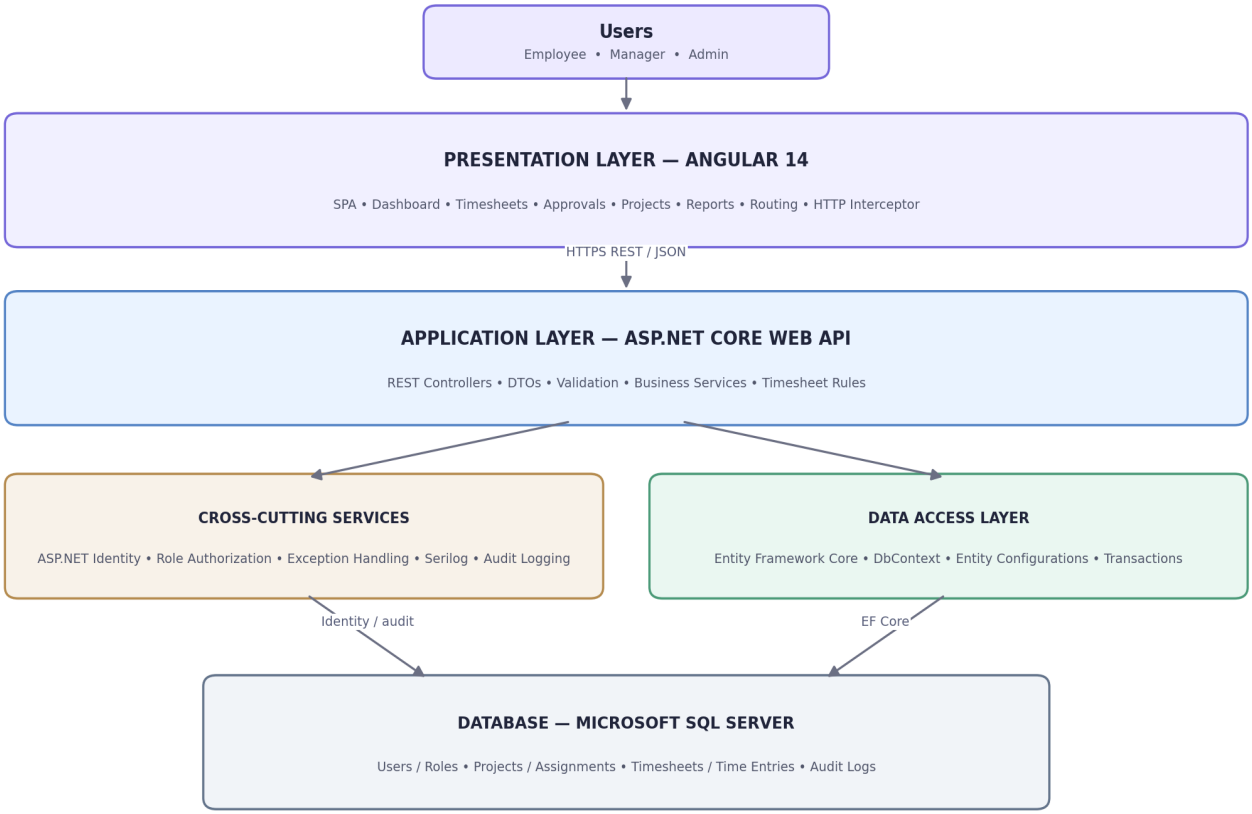
TIMESHEET TRACKER

Solution Architecture & Workflow

Angular 14 | ASP.NET Core Web API | C# | Entity Framework Core | SQL Server

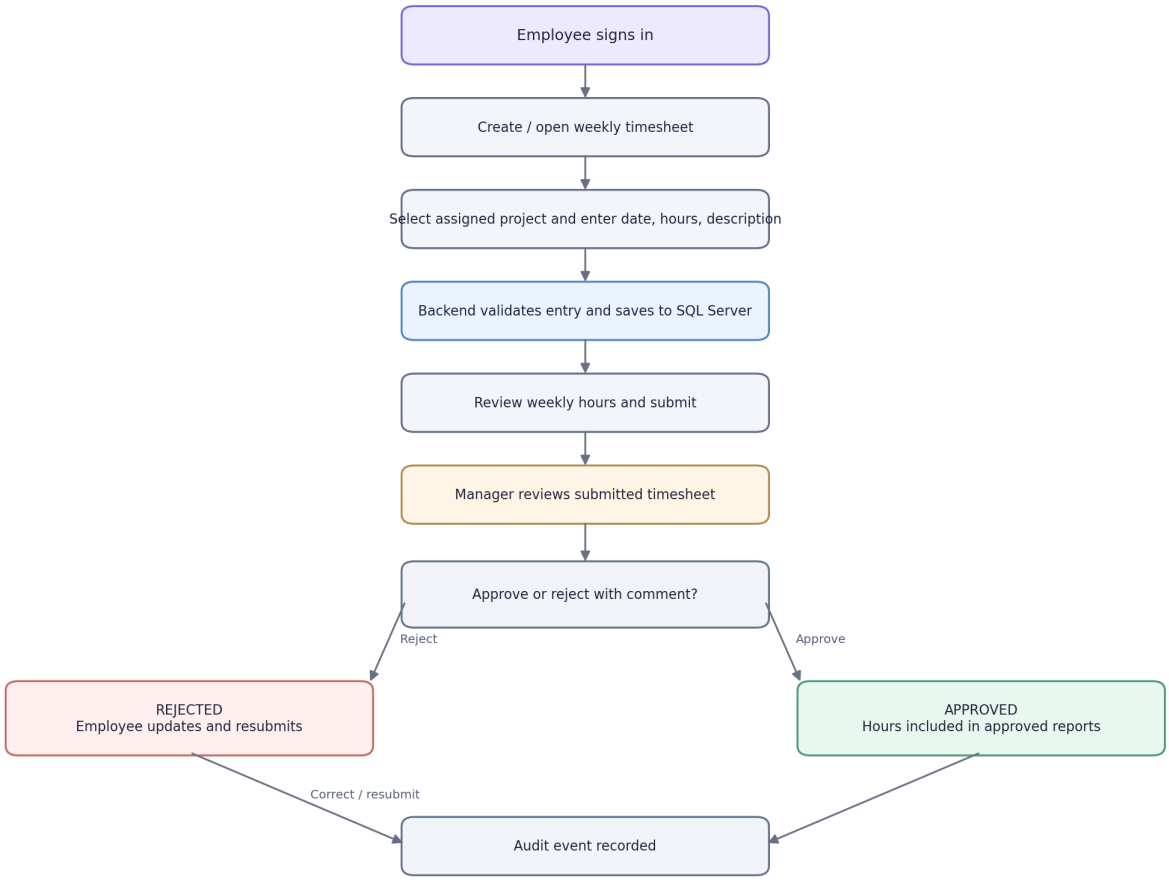
1. High-level solution architecture

Angular 14 is the presentation layer. ASP.NET Core Web API owns business rules, authorization, validation and persistence.



Business rules and authorization are enforced by the backend; Angular is the presentation layer.

2. End-to-end timesheet workflow


Employee signs in
Create / open weekly timesheet
Select assigned project and enter date, hours, description
Backend validates entry and saves to SQL Server
Review weekly hours and submit
Manager reviews submitted timesheet
Approve or reject with comment?
Reject
Approve
REJECTED
Employee updates and resubmits
APPROVED
Hours included in approved reports
Correct / resubmit
Audit event recorded

3. Technology stack

Layer	Technology / responsibility
Frontend	Angular 14.2, TypeScript 4.8, RxJS, Router, Reactive Forms
Backend	C#, ASP.NET Core Web API, REST
Authentication	ASP.NET Core Identity, secure HttpOnly cookies
Data access	Entity Framework Core
Database	Microsoft SQL Server
Cross-cutting	Role authorization, exception handling, Serilog, audit logging

4. Status lifecycle

Draft: Employee can add or update entries.

Submitted: Awaiting manager review.

Approved: Included in approved reporting.

Rejected: Returned with a required comment for correction and resubmission.

5. Component responsibilities

• Angular components render screens and collect input; Angular services call APIs.

• API controllers handle HTTP; DTOs define request/response contracts; business services enforce rules.

• ASP.NET Core Identity authenticates users; authorization checks roles and resource ownership.

• EF Core handles persistence; SQL Server stores users, projects, assignments, timesheets, entries and audit records.

• Serilog and centralized exception handling support diagnostics and consistent error responses.

6. Core data relationships

ApplicationUser 1-to-many Timesheet; Timesheet 1-to-many TimeEntry; Project 1-to-many TimeEntry; ProjectAssignment links ApplicationUser and Project; ApplicationUser 1-to-many AuditLog.

7. API surface

Method	Endpoint	Purpose
POST	/api/auth/login	Sign in
GET	/api/auth/me	Current user
POST	/api/auth/logout	Sign out
GET	/api/timesheets	List sheets
POST	/api/timesheets	Create/get weekly sheet
GET	/api/timesheets/{id}	Get sheet and entries
GET	/api/timesheets/projects	Assigned projects
POST	/api/timesheets/{id}/entries	Add entry
POST	/api/timesheets/{id}/submit	Submit

POST	/api/timesheets/{id}/review	Approve/reject

8. Security and reliability

• Backend role and resource authorization; team-scoped manager access.

• Server-side validation of project assignment, date range, hours, description and status.

• Unique employee/week constraint, transactions and optimistic concurrency.

• Antiforgery protection for cookie-authenticated write requests; HTTPS and secure cookies.

• Audit events, structured logging, automated unit and SQL integration tests.

9. Delivery phases

Foundation and Identity → Employee timesheets → Submission and approvals → Administration → Reports and exports → Testing, security hardening and deployment.

10. Implementation status

This document describes the target architecture. Some capabilities, including manager-to-team authorization, administration screens, live dashboard aggregates, comprehensive automated tests and production security hardening, are planned extensions. Confirm the endpoint contracts against the current source before production approval.